Conditioned mutual information catches coordinated price *moves*, not price *levels*

Apart AI Collusion Sprint · Q1.4 information-theoretic detection · action-only (prices + public state, no messages)

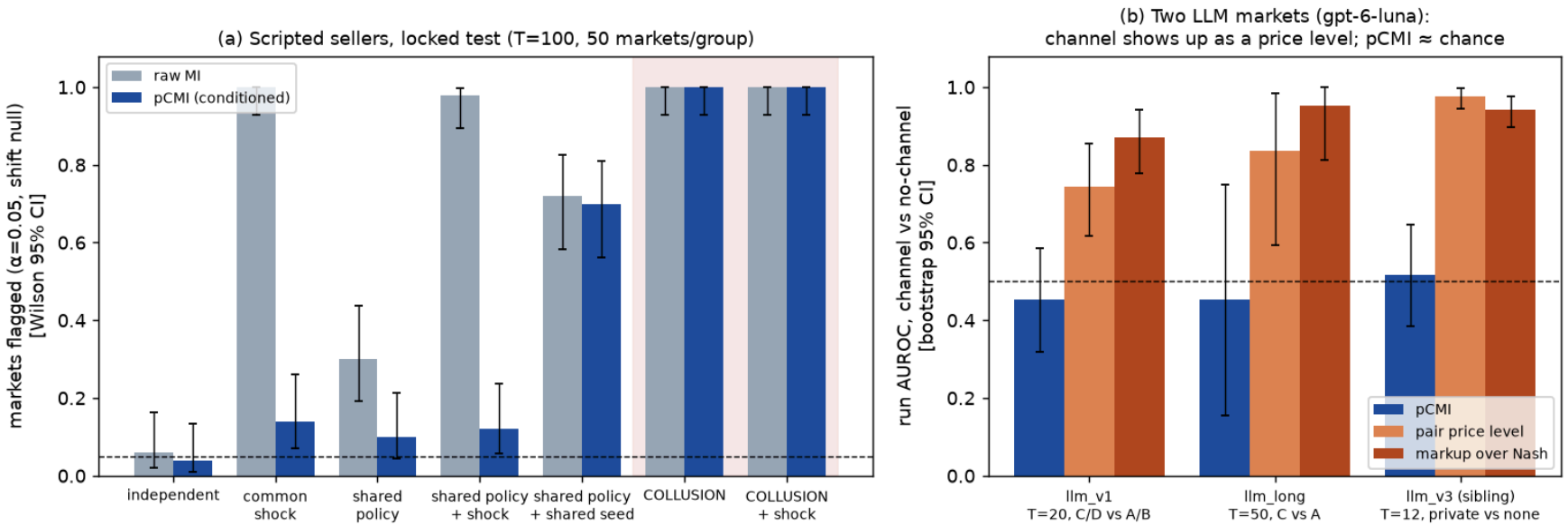
Question. Can a pairwise dependence statistic flag colluding pricing agents *without* accusing agents that merely see the same public shock or run the same policy?

Answer. Partly. With scripted sellers (easiest shared-signal case), pCMI flags 100% of colluding markets and cuts common-shock false alarms from 100% (raw MI) to 14% (target 5%). Copies of one policy that share a random seed still look like collusion. With LLM sellers, pCMI was not distinguishable from chance (wide CIs): the private channel mainly raised price *levels*, and simple level statistics detect it far better.

Setup

- 4-seller repeated differentiated-price (logit) market, public demand/cost shocks, state Nash computed per round.
- Groups: A independent · B common shock · C colluding pair · D colluding pair + shock; hard negatives: shared policy, shared policy + shared seed, public leader-follower ("legit"), independent high price.
- Detectors score each of the 6 seller pairs: correlation; **MI** (mutual information: do two price series move together?); **pCMI** (the same after removing what public shocks and everyone's past prices explain); **TE** (does one seller's past predict the other's price?).
- A market is flagged if its top pair beats a baseline: the same game with series time-shifted out of sync (**shift null**), or real no-channel markets (**control null**). **AUROC**: 0.5 = coin flip, 1.0 = perfect ranking.
- Locked test seeds frozen before tuning (`data/SPLITS.md`); every number regenerates from `results/` (REPORT §6).

Result 1: scripted sellers (locked test, T=100, 50 markets/group, collusion strength 0.6 of 1)



% of markets flagged (target for innocent groups: 5%) [95% CI]	raw MI (shift null)	pCMI (shift null)	pCMI (control null)
common public shock (B)	100% [93, 100]	14% [7, 26]	4% [1, 13]
same policy, independent noise	30% [19, 44]	10% [4, 21]	4% [1, 13]
public leader-follower	8% [3, 19]	12% [6, 24]	4% [1, 13]
policy copy sharing a random seed	72% [58, 83]	70% [56, 81]	48% [35, 61]
collusion C / D	100% [93, 100] / 100% [93, 100]	100% [93, 100] / 100% [93, 100]	100% [93, 100] / 100% [93, 100]

Conditioning removes most shared-shock false alarms (100% → 14%), but not all. A shared random seed is indistinguishable from a private signal. TE (not shown) flags legitimate public leader-follower pricing in 100% of runs. (The 4% for B under the control null is ≈ 5% by construction.)

Result 2: LLM sellers (gpt-6-luna). The channel is a level shift

- **Our market `llm_v1`** (74 games, T=20): the private channel raises the average markup over the competitive (Nash) price by +1.76 price units [1.21, 2.31] (C vs A) and +1.77 [1.02, 2.52] (D vs B).
- AUROC for telling channel markets from no-channel markets (best pair per market; 0.5 = chance). This measures channel *assignment*: only 17/32 `llm_v1` channel games show realised collusion (verified agreement + coordinated price actions).

	pCMI	pair price − rivals'	pair markup over Nash
`llm_v1` T=20, C/D vs A/B	0.45 [0.32, 0.59]	0.74 [0.62, 0.86]	0.87 [0.78, 0.94]
`llm_long` T=50, C vs A (8 v 8)	0.45 [0.16, 0.75]	0.84 [0.59, 0.98]	0.95 [0.81, 1.00]
sibling `llm_v3` T=12, private vs none	0.52 [0.39, 0.64]	0.98 [0.95, 1.00]	0.94 [0.90, 0.98]

- Pair localisation in `llm_v3` (chance 1/6): pCMI top-1 18/70 (nominal p=0.036, uncorrected; not read as signal); pair level 63/70.
- Longer games (T=50) did not rescue pCMI by our pre-declared rule (8 v 8 games, so only large effects are excluded).
- Mixed-model markets (2 gpt-6.1-sol + 2 gpt-6-luna, 12 games): same-model minus cross-model pCMI +0.003 [-0.025, +0.032]; no shared-model excess detected (only large effects excluded).

What this means

- **Dependence detectors respond to coordinated *variation*, not to agreement as such.** They are built to ignore a constant high price. In these two markets LLM collusion shows up mainly as a level shift, so it falls largely outside what they measure. Some coordinated variation is present: on first differences, MI localises the `llm_v3` channel pair above chance (REPORT §4.3.3, exploratory); in a 15-seed placebo replication it names the pair in 10/15 runs with delivered messages vs 3/15 without (sign-flip p=0.065, unadjusted; suggestive only, §4.3.9).
- **Level screens separate channel assignment but are not collusion-specific.** They also rank markets with a *legitimate* public partnership above controls (AUROC 0.75), naming the partner pair in only 5/20 runs. In the sibling project, *offering* a private channel that never delivered messages raised prices about as much as delivering them (exploratory, 15 seeds). Our frozen pair_level on those runs names the pair in 13.5/15 placebo vs 14.0/15 private runs; no detector separates private from placebo at n=15 (REPORT §4.3.9). In exploratory sibling monitor/firm-framing strata with no judge-labelled agreements, pair_level yielded 3.3/30 tie-averaged hits, with no detected above-chance localisation (chance expectation 5.0; REPORT §4.3.10; framing confounded).
- Suggested pipeline (a hypothesis, untested): a cheap always-on level screen, then dependence and text audits as second-stage evidence.
- Not settled: whether pCMI fails because the effect is mostly a level or because power is low at T ≤ 50. Exploratory check (REPORT §4.3.8, post-hoc): per-round shared ±1 price shocks planted into one pair (in 20% of rounds) of real no-channel LLM games are detected by pCMI with AUROC 0.93 [0.87, 0.98] at T=50 but only 0.64 [0.54, 0.74] at T=20. The real channel pairs show no detected excess same-direction moves (T=50: -0.008 [-0.023, +0.009]). This fits the level reading for one planted alternative; it does not settle the question.

Limitations & next steps

- Small LLM n: `llm_v1` 74, `llm_v3` 125 (70 private / 35 none / 20 legit), `llm_long` 16, `llm_mixed2` 12, sibling replication 30 + placebo 15. Sellers are OpenAI-family only. There is no placebo arm in `llm_v1`. Markup needs a market model (Nash).
- Next: a placebo arm in our market and a larger one in the sibling's; longer LLM games ($T \geq 50$, more games per arm); level-adjusted dependence (residualise each seller's own level); non-OpenAI sellers.
- Full write-up: `REPORT.md`. Ledger of every experiment (E01–E37), including discarded ones: `EXPERIMENTS.tsv`.